BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, AJK, MIRPUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 21/01

Retyped faithfully from the original board paper — same questions, no changes.

MATHEMATICS (PART –I) (OBJECTIVE PART)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 21/01

Note: Note:- Write your Roll No. in space provided. Over writing, cutting, using of lead pencil will result in loss of marks. All questions are to be attempted.

Q1. (Marks (20)) Each question has four possible answers, Tick (√) the correct answer.

Q1. Solution set of the equation $x^2 + 7x + 10 = 0$ is;

(A) $\{-2,5\}$ (B) $\{-2,-5\}$

(C) $\{2,-5\}$ (D) $\{2,5\}$

Q2. Partial fractions of $3x/((x-1)(x+2))$ are;

(A) $1/(x-1) + 2/(x+2)$ (B) $2/(x-1) + 1/(x+2)$

(C) $1/(x-1) - 2/(x+2)$ (D) $2/(x-1) - 1/(x+2)$

Q3. A.M between $x+1$ and $(x-1)$ is;

(A) $x-1$ (B) x

(C) $(x+1)/2$ (D) $(x-1)/2$

Q4. The common ratio of a geometric sequence cannot be;

(A) 3 (B) 1

(C) 2 (D) 0

Q5. An infinite geometric series is convergent if;

(A) $|r| \leq 1$ (B) $|r| < 1$

(C) $|r| < 2$ (D) $|r| > 1$

Q6. Trichotomy is the property of;

(A) Inequality
(B) Equality
(C) Division
(D) Subtraction

Q7. Converse of $p \rightarrow q$ is;

(A) $\sim p \rightarrow q$
(B) $p \rightarrow \sim q$
(C) $q \rightarrow p$
(D) $\sim p \rightarrow \sim q$

Q8. If a system of linear equations has a unique solution or infinitly many solutions then it can be known as;

(A) Consistent system
(B) Inconsistent system
(C) Non-linear system
(D) Unique solution

Q9. The matrix [[5,0,0],[0,5,0],[0,0,5]] is a;

(A) Unit matrix
(B) Scalar Matrix
(C) Zero matrix
(D) Rectangular matrix

Q10. If a polynomial function f(x) is divided by $(x-a)$ then f(x) is called;

(A) Divisor
(B) Quotient
(C) Remainder
(D) Dividend

> Note: (P-T-O)

> Note: Page 2

Q11. The period of tan(x/3) is;

(A) π
(B) 2π
(C) 3π
(D) 4π

Q12. Number of the elements of a triangle are;

(A) 3
(B) 4
(C) 6
(D) 8

Q13. In-radius r of a triangle is;

(A) $S\Delta$
(B) Δ/S
(C) S/Δ
(D) $4ab/\Delta$

Q14. $\text{Cos}(\tan^{-1}(\sqrt{3}))$ is equal to;

(A) $1/2$
(B) $-1/2$
(C) $\sqrt{3}/2$
(D) $-\sqrt{3}/2$

Q15. If Cos(2x) = 0 then solution in 1st quadrant is;

(A) 30° (B) 45°

(C) 60° (D) 15°

Q16. 8!/6! =

(A) 8 (B) 6

(C) 50 (D) 56

Q17. The expansion $(1+x)^{-3}$ holds when;

(A) $|x| > 0$ (B) $|x| < 1$

(C) $|x| \leq 1$ (D) $|x| < 0$

Q18. Cos(330°) = _____

(A) 1/2 (B) √3/2

(C) 2/√3 (D) 1/√2

Q19. If l and r are in Cms, then the unit of θ is in;

(A) Radians (B) Degree

(C) Cm^2 (D) Cm

Q20. Sin(2π − θ) = _____

(A) Sinθ (B) −Sinθ

(C) Cosθ (D) −Cosθ

Note: (The End)

MATHEMATICS PAPER : PART – I (SUBJECTIVE PART)

Time Allowed: 2:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 21/01

Note: Note:- Attempt any TWENTY FIVE (25) short questions in all selecting eight from Q. 2 and Q. 3 each and nine from Q. 4. (25 x 2 = 50)

Note: SECTION – I

Q2. (Marks (2 x 8 = 16)) Write short answers of any eight questions.

(i) Show that $\forall z_1, z_2 \in C$, conjugate($z_1 z_2$) = conjugate(z_1) conjugate(z_2)

(ii) Find out real and imaginary parts of complex number $(\sqrt{3} + i)^3$

(iii) Factorize $a^2 + 4b^2$

(iv) For the set A = {1,2,3,4} Find the relation {(x,y) | x + y > 5}. Also state domain and range of this relation.

(v) Prepare the table of addition of the elements of set of residue classes modulo 4.

(vi) Use Venn diagram to verify $(A-B)^c \cap B = B$

(vii) Solve the matrix equation for X, $2X - 3A = B$. If A = [[1,−1,2],[−2,4,5]], B = [[3,−1,0], [4,2,1]]

(viii) Without expansion show that |1 2 3; 4 5 6; 7 8 9| = 0

(ix) If A = [[1,−2,3],[−2,3,1],[4,−3,2]] Find A_{12} and A_{32}

(x) Solve x^(1/2) − x^(1/4) − 6 = 0

(xi) Find four fourth roots of 625.

(xii) If α, β are the roots of $3x^3 - 2x + 4 = 0$ Find the value of α/β + β/α

Q3. (Marks (2 x 8 = 16)) Write short answers of any eight questions.

(i) Write 1/((x−1)(2x−1)(3x−1)) in the form of partial fraction without finding constants.

(ii) If (7x+25)/((x+3)(x+4)) = A/(x+3) + B/(x+4), then find the value of B.

(iii) If 1/a, 1/b and 1/c are in G.P. Show that the common ratio is ±√(a/c)

(iv) Fine the 5the term of G.P. 3,6,12,...

(v) Find A.M. between x−3 and x+5

(vi) If $a_{n-3} = 2n - 5$ Find the n th term of the sequence.

(vii) What is the probability that a slip of numbers divisible by 4 is picked from the slips bearing numbers 1,2,3,...,10?

(viii) Find the value of n if ${}^nC_{12} = {}^nC_6$

(ix) Evaluate: 9!/(2!(9−2)!)

(x) Verify the statement $1+2+4+...+2^{n-1} = 2^n-1$ for n=2 and n=3

(xi) If x is so small that it square and higher powers can be neglected, then show that (1−x)/√(1+x) ≈ 1 − (3/2)x

(xii) Find the term involving x^{-2} in the expansion of $(x - 2/x^2)^{13}$

> Note: P.T.O

> Note: Page 2

Q4. (Marks (2 x 9 = 18)) Write short answers of any nine questions.

(i) Convert into radians the 54°45′

(ii) Find the value of Cosθ and tanθ, when Sinθ = 12/13 and terminal arm of the angle is in quadrant I

(iii) Prove that $\frac{\sin\theta}{1+\cos\theta} + \cot\theta = \operatorname{cosec}\theta$

(iv) Without use of table or calculator, Find the value of $\sin(-300°)$

(v) Prove that $\tan(45°+A)\tan(45°-A) = 1$

(vi) Express $\sin 5x + \sin 7x$ as a product.

(vii) Find the period of $\tan 4x$

(viii) Find the measure of greatest angle, if sides of the triangle are 16,20,33.

(ix) Find the area of the triangle, whose sides are a=18, b=24, c=30

(x) At the top of a cliff 80 m high, the angle of depression of a boat is 12°. How for is the boat from the cliff?

(xi) Show that $\cos^{-1}(12/13) = \sin^{-1}(5/13)$ where domain of the function is $[0,\pi]$

(xii) Find the value of the equation, which lies in $[0,2\pi]$, $\sec x = -2$

(xiii) Solve the trigonometric equation $\tan^2\theta = 1/3$

> Note: SECTION – II

> Note: Note:- Attempt any three questions. (10 x 3 = 30)

Q5. (Marks (10)) (a) Show that $|x\ 1\ 1\ 1;\ 1\ x\ 1\ 1;\ 1\ 1\ x\ 1;\ 1\ 1\ 1\ x| = (x+3)(x-1)^3$ (05)

Q. (Marks (05)) (b) Solve the equation $3\hat{}(2x-1) + 12.3^x + 81 = 0$

Q6. (Marks (10)) (a) Resolve into partial fraction. $1/((x-1)^2(x^2+2))$ (05)

Q. (Marks (05)) (b) Find the 11th term of the sequence. $1 + i, 2, 4/(1+i), \ldots$

Q7. (Marks (10)) (a) A die is thrown twice. What is the probability that the sum of the number of dots shown is 3 or 11. (05)

Q. (Marks (05)) (b) Find the term independent of x in the expansion $(\sqrt{x} + 1/(2x^2))^{10}$

Q8. (Marks (10)) (a) Without using calculator / tables, show that $\cos 20°\cos 40°\cos 80° = 1/8$ (05)

Q. (Marks (05)) (b) If $\sin\theta = 12/13$ and terminal arm of the angle is in quadrant I, then find the remaining Trigonometric functions.

Q9. (Marks (10)) (a) Solve the following triangle using first law of tangents and then law of sines. If b = 14.8, c = 16.1 and $\alpha = 42°45'$ (05)

Q. (Marks (05)) (b) Prove that $2\tan^{-1}(2/3) = \sin^{-1}(12/13)$

> Note: (The End)